

Week 4 Discussion: Hedging with Futures & Interest Rates (Lec 8–10)

Main focus: basis risk & cross-hedging (min-variance hedge ratio), and interest-rate basics (compounding, zero rates, bond pricing/yield, forward rates).

1 Rapid Review

Module A) Static hedges with futures + basis risk

Notation: hedge set at t_1 , closed at t_2 ; spot S_t (asset to hedge), futures F_t (contract used), basis $b_t := S_t - F_t$.

- **Short hedge (selling later):** short futures at t_1 , close out at t_2 .

$$\text{Effective selling price} = S_{t_2} + (F_{t_1} - F_{t_2}) = F_{t_1} + (S_{t_2} - F_{t_2}) = F_{t_1} + b_{t_2}.$$

- **Long hedge (buying later):** long futures at t_1 , close out at t_2 .

$$\text{Effective buying price} = S_{t_2} - (F_{t_2} - F_{t_1}) = F_{t_1} + b_{t_2}.$$

- **Basis risk:** at t_1 , F_{t_1} known but b_{t_2} unknown. Perfect hedge $\Leftrightarrow b_{t_2} \approx 0$ (same underlying + close to expiry).

Module B) Cross-hedging & minimum-variance hedge ratio (ignore daily settlement)

Cross-hedging: hedge asset S using futures on a related asset F .

Let $\Delta S = S_{t_2} - S_{t_1}$, $\Delta F = F_{t_2} - F_{t_1}$ with s.d. σ_S, σ_F and corr. ρ . If N_A = exposure size (units of asset to hedge) and each futures contract is Q_F units, then:

$$h^* = \rho \frac{\sigma_S}{\sigma_F}, \quad N^* = \frac{h^* N_A}{Q_F}.$$

Direction: if you own the asset and fear price falling $\Rightarrow$ short futures; if you need to buy later and fear price rising $\Rightarrow$ long futures.

Module C) Interest rates & bonds (continuous compounding in this class)

- **Compounding:** $A \mapsto A(1 + r/m)^{mn}$ (m-times/year), or $A \mapsto Ae^{r_c n}$ (continuous).

$$r_c = m \ln\left(1 + \frac{r_m}{m}\right), \quad r_m = m\left(e^{r_c/m} - 1\right).$$

- **Discounting:** PV of A received at t is Ae^{-rt} (continuous).
- **Zero rate (spot rate):** $R(t) : DF(0, t) = e^{-R(t)t}$.
The spot interest rate for a specific maturity time t . It measures the opportunity cost of capital for a **single payment** occurring at time t . (Serve as a baseline)
- **Bond price (spot curve):** $B_0 = \sum_i c_i e^{-R(t_i)t_i}$.
Theoretical price found by discounting each cash flow c_i using its **specific** time's rate $R(t_i)$.
- **Bond yield (single rate):** $y : B_0 = \sum_i c_i e^{-yt_i}$ (solve numerically).
The single constant "average" rate that equates the PV of all cash flows to the market price B_0 .

2 Practice Problems

P1. (3.13) Optimal hedge ratio for live cattle (minimum variance).

The standard deviation of monthly changes in the **spot** price of live cattle is 1.2 (cents/lb). The standard deviation of monthly changes in the **futures** price of live cattle for the closest contract is 1.4 (cents/lb). The correlation between futures price changes and spot price changes is 0.7. It is now October 15. A producer is committed to **purchasing** 200,000 pounds of live cattle on November 15. The producer wants to use

December live cattle futures to hedge risk. Each futures contract is for 40,000 pounds.

- Compute the minimum-variance hedge ratio h^* .
- Should the producer go **long** or **short** futures? (one line)
- How many futures contracts should be used? (round to nearest integer)
- Describe the strategy in words (when to enter and when to close out).

TA Key for P1

- $h^* = \rho \sigma_S / \sigma_F = 0.7 \times (1.2/1.4) = 0.7 \times 0.8571429 = 0.60$.
- They will **buy** later and fear price rising $\Rightarrow$ **long** futures.
- $N^* = \frac{h^* N_A}{Q_F} = \frac{0.60 \times 200,000}{40,000} = 3.0 \Rightarrow \boxed{3 \text{ contracts}}$.
- On Oct 15, **go long 3 December live cattle futures**; on Nov 15 (when buying cattle in spot market), **close out** the futures position by taking the opposite side (sell 3 contracts).

P2. (4.2) Bond price from yield; back out the 18-month zero rate (semiannual comp.).

The 6-month and 1-year **zero rates** are both 5% per annum. A bond has a life of 18 months and pays a coupon of 4% per annum (with **semiannual** payments and one having just been made). The **yield** is 5.2% per annum. All rates are quoted with **semiannual compounding**. Assume face value is \$100.

- Using the yield (single discount rate), compute the bond price.
- Using the given 6-month and 1-year zero rates, infer the **18-month zero rate**.

TA Key for P2

Cashflows (coupon just paid at $t = 0$): 2 at $t = 0.5$, 2 at $t = 1.0$, and 102 at $t = 1.5$ (since coupon rate 4% p.a. with semiannual coupon = $100 \times 0.04/2 = 2$).

- Yield $y = 5.2\%$ (semiannual) $\Rightarrow$ per half-year rate $y/2 = 0.026$.

$$B_0 = \frac{2}{(1.026)^1} + \frac{2}{(1.026)^2} + \frac{102}{(1.026)^3} \approx 98.29.$$

- Zero rates $R(0.5) = R(1.0) = 5\%$ (semiannual) $\Rightarrow$

$$DF(0, 0.5) = \frac{1}{(1.025)^1}, \quad DF(0, 1.0) = \frac{1}{(1.025)^2}.$$

Let $DF(0, 1.5)$ be the 18-month discount factor. Then

$$98.2897 = 2 DF(0, 0.5) + 2 DF(0, 1.0) + 102 DF(0, 1.5),$$

so

$$DF(0, 1.5) = \frac{98.2897 - 2/(1.025) - 2/(1.025)^2}{102} \approx 0.92583.$$

Now set $DF(0, 1.5) = \frac{1}{(1 + R_{1.5}/2)^3}$ (3 half-year periods), giving

$$1 + \frac{R_{1.5}}{2} = \left(\frac{1}{DF(0, 1.5)} \right)^{1/3} \Rightarrow R_{1.5} = 2 \left(DF(0, 1.5)^{-1/3} - 1 \right) \approx 5.204\%.$$

So the 18-month zero rate is $\boxed{R_{1.5} \approx 5.20\% \text{ p.a. (semiannual comp.)}}$.